

Ocular AI — User Manual

A comprehensive guide to system initialization, real-time spatial mapping, and multi-sensory feedback interface utilization.

✂ Getting Started

First-Time Setup

1. **Install the App:** Download and install the `OcularAI.apk` file directly onto your Android device.
2. **Launch System:** Open the **Ocular AI** application from your device drawer.
3. **Grant Camera Permission:** Allow camera access when prompted (Strictly required for real-time edge frame object analysis).
4. **Grant Microphone Permission:** Allow microphone access when prompted (Required for processing device-side localized voice commands).
5. **Optimal Positioning:** Hold your device steadily at chest height with the rear primary camera lens facing directly forward.
6. **System Confirmation:** The application will synthesize an auditory phrase stating "**Ocular AI ready**" once all tracking engines instantiate.

📱 Operation & Navigation

Basic Navigation Workflow

- **Orientation:** Maintain your smartphone strictly in vertical (**Portrait Mode**) configuration.
- **Directional Alignment:** Point the device's rear camera directly toward your intended path of movement.
- **Auditory Cues:** Keep your device volume up or use headphones to safely listen to continuous real-time obstacle updates.

- **Tactile Sensing:** Pay attention to rhythmic haptic vibration patterns to identify critical environmental hazards without looking at the display.

Understanding Auditory Announcements

The localized Text-to-Speech (TTS) engine synthesizes situational alerts using a standardized priority format:

"[Object Name] on your [Directional Vector], [Relative Distance Status]"

- *Example:* "Chair on your left, near."
- *Example:* "Person on your center, approaching."

Voice Command Matrix

To interact with the assistive voice interface, tap the high-visibility **Voice Button** located at the bottom center of the HUD overlay, wait for the listening cue, and speak one of the following exact command strings:

Voice Command	System Action / Response
"What's around me?"	Initiates an immediate verbal summary of all current prioritized objects in the tracking queue.
"Repeat"	Re-announces the exact last visual obstacle update saved in the telemetry log.

Voice Command	System Action / Response
"Emergency"	Triggers an instantaneous local safety broadcast sequence and sets haptic warnings to maximum intensity.
"Silence"	Suspends the active Text-to-Speech audio announcement stream (Mutes voice output).
"Resume"	Re-enables the Text-to-Speech speech synthesizer module (Unmutes voice output).

⚡ Environmental Alert Triggers

The integrated `UIController` and `EnvironmentAnalyzer` automatically parse environmental degradation factors. Pay close attention to these explicit system states:

Auditory / UI Alert	Physical Meaning	Recommended Safety Action
"Person ahead, near"	An individual is occupying your direct vertical lane closely.	Proceed with caution and prepare to step aside.

Auditory / UI Alert	Physical Meaning	Recommended Safety Action
"Warning! Car approaching"	A motor vehicle is detected within your frontal plane closing distance.	Stop moving immediately and verify your surroundings.
"Low light condition"	Ambient pixel luminosity values have dropped below safe processing levels.	Decrease your walking pace or move to a illuminated area.
"Lens blocked"	The primary camera feed has lost structural visibility entirely.	Check for lens obstruction or clean the hardware camera glass.

Haptic Vibration Signature Patterns

Ocular AI uses customized, discrete haptic motor patterns to communicate perimeter data silently without requiring structural screen observation:

- **Single Short Tap:** A general background static object has entered the outer scanning radius.
- **Quick Double-Tap:** A moving human presence (Person) has been successfully verified by the neural network.
- **Strong Double-Pulse: Immediate Hazard Flag.** An obstacle is directly obstructing your immediate line of travel.
- **Long Double-Rumble:** A vehicle or high-mass movement profile is detected nearby.

- **Continuous Triple Pulse:** The emergency state has been activated.

💡 Pro-Tips for Peak System Performance

- **Lens Maintenance:** Wipe down your device's camera lens regularly. Smudges degrade the classification accuracy of the `efficientdet_lite0` network.
- **Stability First:** Avoid erratic swinging motions. Keep the device firmly planted at chest height to allow the frame processing interval loops to build tracking stability.
- **Power Management:** Continuous live camera rendering and localized machine learning inference draw significant battery power. Keep your phone charged before long navigation sessions.
- **Hands-Free Setup:** For optimal everyday utility, use a secure chest mount or a phone lanyard to maintain consistent forward alignment while keeping your hands free.